Temperature_Analysis

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```r
rm(list = ls())
```

```r
knitr::opts_chunk$set(cache = TRUE, cache.lazy = FALSE, warning = FALSE,
                      message = FALSE,echo = TRUE, dpi = 300,
                      fig.width = 15, fig.height = 10)
```

Reading the data from the directory

```r
## Daily data
data <- readRDS(file = "/Users/mandarphatak/Climate-Analysis/DWD_Folder/Data_DWD/Hamburg_Fuhlsbüttel.rds
data <- data |> dplyr::filter(date < "2025-01-01")
```

```r
library(tidyverse)
library(SPEI)
library(drought)
library(extRemes)
library(heatwaveR)
```

data tranformation :- We select the te

```
##       Tavg            Tmax            Tmin            prec
##  Min.   : 6.794   Min.   :10.88   Min.   :2.556   Min.   : 392.3
##  1st Qu.: 8.352   1st Qu.:12.08   1st Qu.:4.593   1st Qu.: 681.2
##  Median : 9.140   Median :12.92   Median :5.099   Median : 742.9
##  Mean   : 9.060   Mean   :12.91   Mean   :5.137   Mean   : 759.4
##  3rd Qu.: 9.739   3rd Qu.:13.59   3rd Qu.:5.891   3rd Qu.: 831.2
##  Max.   :11.180   Max.   :15.07   Max.   :7.161   Max.   :1071.8
##     rel_hum
##  Min.   :69.43
##  1st Qu.:78.50
##  Median :80.03
##  Mean   :79.77
##  3rd Qu.:81.24
##  Max.   :84.81
```

Focusing on Temperature data

Data Over View :- We perform the following data manipulation for desired data overview

```
annual_temperature_tbl_long$Temperature_Type <- factor(
  annual_temperature_tbl_long$Temperature_Type,
  levels = c("Tmax", "Tavg", "Tmin"))


trends_tbl <- annual_temperature_tbl_long %>%
  group_by(Temperature_Type) %>%
  summarize(trend = coef(lm(Temperature ~ year))[2]) %>%
  ungroup()


trends_tbl <- trends_tbl %>%
  mutate(
    label = case_when(
      Temperature_Type == "Tmax" ~ paste0("Maximum Temperature (Trend: +", round(trend, 4), "°C/year)")
      Temperature_Type == "Tavg" ~ paste0("Average Temperature (Trend: +", round(trend, 4), "°C/year)")
      Temperature_Type == "Tmin" ~ paste0("Minimum Temperature (Trend: +", round(trend, 4), "°C/year)")
    ),
    color = case_when(
      Temperature_Type == "Tmax" ~ "#ff1744",
      Temperature_Type == "Tavg" ~ "#fc8d59",
      Temperature_Type == "Tmin" ~ "#4575b4"
    ),
    y = case_when(  # adjust based on your plot's Y-axis ranges
      Temperature_Type == "Tmax" ~ 15,
      Temperature_Type == "Tavg" ~ 10.55,
      Temperature_Type == "Tmin" ~ 6.5
    ),
    x = 1938  # horizontal position of the labels
  )
```

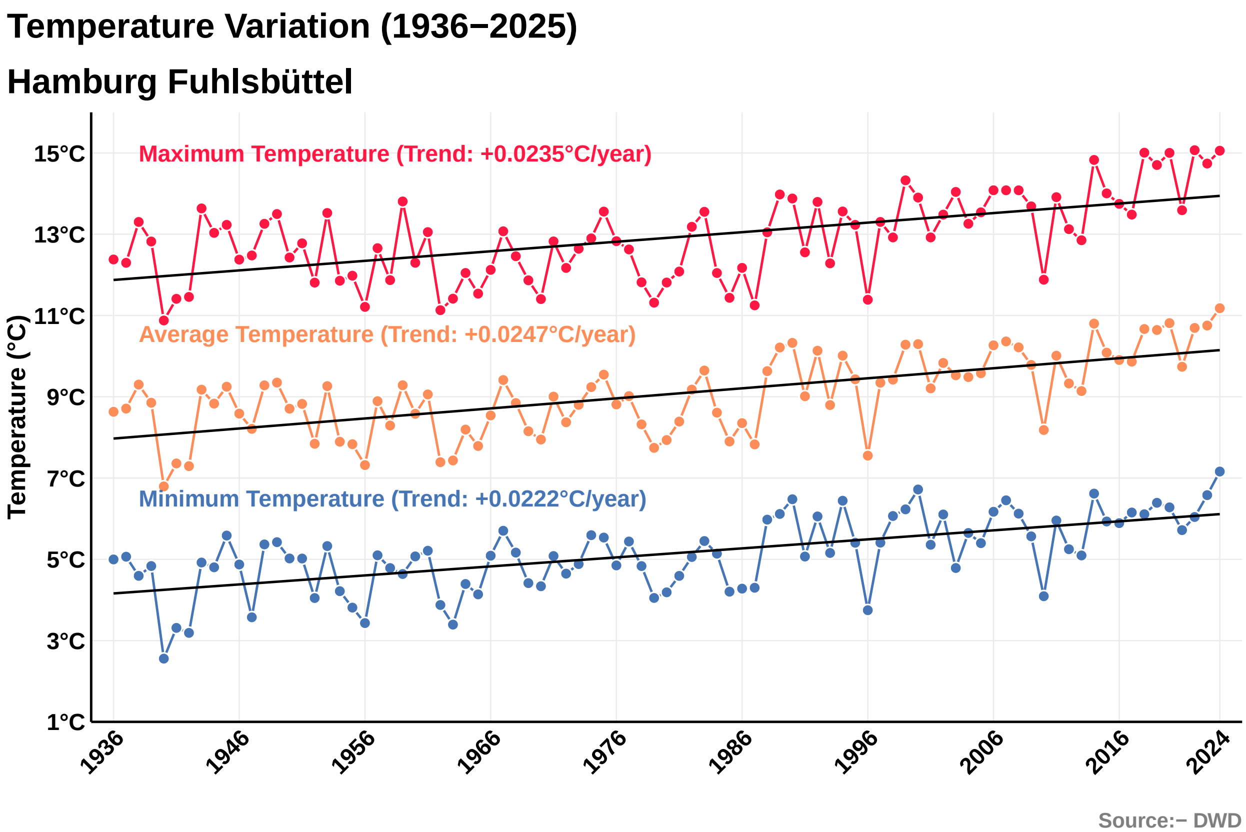
Temperature Variation

Overview of Temperature Variation 1936-2024

```
ggplot(annual_temperature_tbl_long, aes(x = year, y = Temperature, colour = Temperature_Type)) +
  geom_line(linewidth = 1) +
  geom_point(aes(fill = Temperature_Type), shape = 21, size = 4, stroke = 1, colour = "white") +
  # Add separate trend lines for each temperature type
  geom_smooth(aes(group = Temperature_Type), method = "lm", se = FALSE, linewidth = 1,
              show.legend = FALSE, color = "black") +
  geom_text(
    data = trends_tbl,
    aes(x = x, y = y, label = label, color = Temperature_Type),
```

```r
  fontface = "bold", hjust = 0, size = 7, show.legend = FALSE
) +

# Fixed color scales with intuitive temperature colors
scale_color_manual(
  values = c("Tmax" = "#ff1744",    # Hot red for maximum
             "Tavg" = "#fc8d59",    # Orange for average
             "Tmin" = "#4575b4"),   # Cool blue for minimum
  labels = c("Tmax" = "Maximum Temperature",
             "Tavg" = "Average Temperature",
             "Tmin" = "Minimum Temperature"),
  name = "Temperature Type"
) +
scale_fill_manual(
  values = c("Tmax" = "#ff1744",    # Same colors for consistency
             "Tavg" = "#fc8d59",
             "Tmin" = "#4575b4"),
  guide = "none"  # Hide fill legend since color legend shows the same info
) +

scale_x_continuous(breaks = c(seq(1936, 2024, 10), 2024),
                   expand = c(0.02, 0.02),
                   limits = c(1936, 2024)) +
scale_y_continuous(
  limits = c(1, 16),
  breaks = seq(1, 16, by = 2),
  labels = function(x) paste0(format(x, nsmall = 0), "°C"),
  expand = c(0, 0)
) +

labs(
  title = "Temperature Variation (1936-2025)",
  subtitle = "Hamburg Fuhlsbüttel",
  caption = "Source:- DWD",
  x = "",
  y = "Temperature (°C)"
) +
theme_minimal() +
theme(
  plot.title.position = "plot",
  plot.title = element_text(hjust = 0, face = "bold", colour = "black", size = 30, margin = margin(b =
  plot.subtitle = element_text(hjust = 0, face = "bold", colour = "black", size = 30, margin = margin
  plot.caption = element_text(size = 18, face = "bold", color = "gray50", hjust = 1),  #
  axis.line = element_line(color = "black", linewidth = 1),
  axis.title = element_text(face = "bold", color = "black", size = 22),
  axis.ticks = element_blank(),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(linewidth = 0.5),
  axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", color = "black", size = 20),
  axis.text.y = element_text(face = "bold", color = "black", size = 20),
  legend.text = element_text(size = 18, face = "bold"),
  legend.title = element_text(size = 18, face = "bold"),
  legend.position = "bottom",  # Better positioning for temperature legend
```

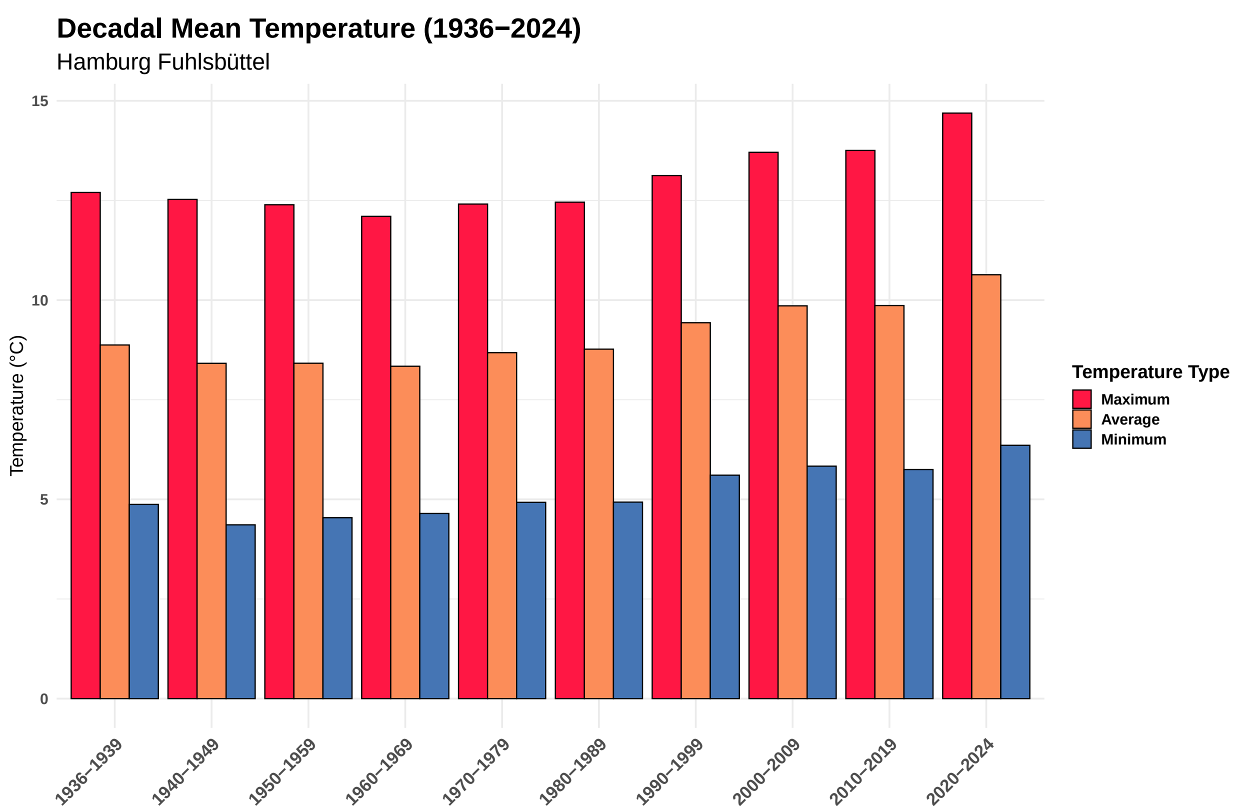
```r
    panel.background = element_blank(),
    plot.background = element_blank()
  ) +
   guides(color = "none", fill = "none")
```



Decadal Changes in Temperature

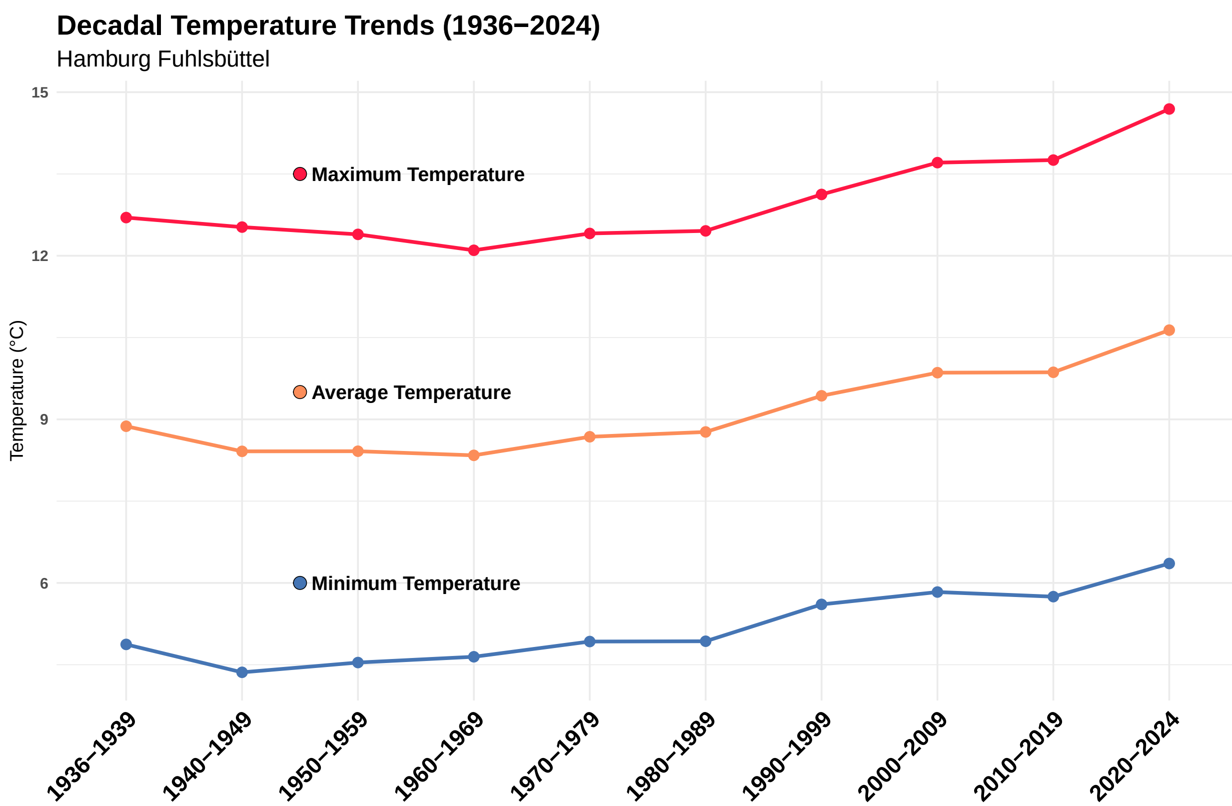
```r
 annual_temperature_tbl_long <- annual_temperature_tbl_long |>
  mutate(decade_group = case_when(
    year >= 1936 & year <= 1939 ~ "1936-1939",
    year >= 1940 & year <= 1949 ~ "1940-1949",
    year >= 1950 & year <= 1959 ~ "1950-1959",
    year >= 1960 & year <= 1969 ~ "1960-1969",
    year >= 1970 & year <= 1979 ~ "1970-1979",
    year >= 1980 & year <= 1989 ~ "1980-1989",
    year >= 1990 & year <= 1999 ~ "1990-1999",
    year >= 2000 & year <= 2009 ~ "2000-2009",
    year >= 2010 & year <= 2019 ~ "2010-2019",
    year >= 2020 & year <= 2024 ~ "2020-2024",
    TRUE ~ NA_character_
  ))

  decadal_summary_tbl <- annual_temperature_tbl_long %>%
  group_by(decade_group, Temperature_Type) %>%
  summarise(decadal_mean = mean(Temperature, na.rm = TRUE), .groups = "drop")

library(ggplot2)
```

```r
ggplot(decadal_summary_tbl, aes(x = decade_group, y = decadal_mean, fill = Temperature_Type)) +
  geom_bar(stat = "identity", position = "dodge", color = "black") +
  scale_fill_manual(
    values = c("Tmax" = "#ff1744", "Tavg" = "#fc8d59", "Tmin" = "#4575b4"),
    name = "Temperature Type",
    labels = c("Tmax" = "Maximum", "Tavg" = "Average", "Tmin" = "Minimum")
  ) +
  labs(
    title = "Decadal Mean Temperature (1936-2024)",
    subtitle = "Hamburg Fuhlsbüttel",
    x = "",
    y = "Temperature (°C)"
  ) +
  theme_minimal(base_size = 16) +
  theme(
    plot.title = element_text(face = "bold", size = 24),
    plot.subtitle = element_text(size = 20),
    axis.text.x = element_text(angle = 45, hjust = 1, face = "bold",size = 15),
    axis.text.y = element_text(face = "bold"),
    legend.title = element_text(face = "bold"),
    legend.text = element_text(face = "bold")
  )
```



Variation in Temperature across decade

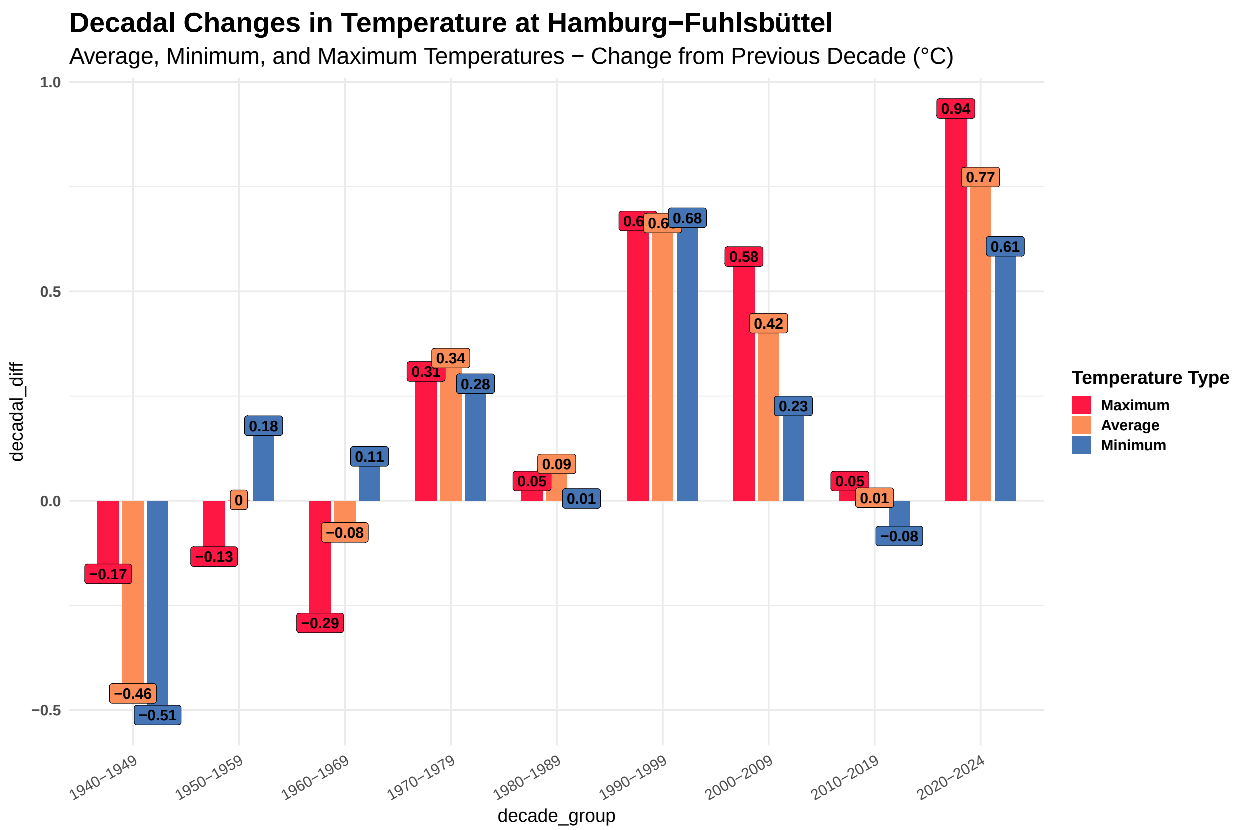
```
ggplot(decadal_summary_tbl, aes(x = decade_group, y = decadal_mean, group = Temperature_Type, color = Te
  geom_line(linewidth = 1.5) +
  geom_point(size = 4) +
  scale_color_manual(
    values = c("Tmax" = "#ff1744", "Tavg" = "#fc8d59", "Tmin" = "#4575b4"),
    name = "Temperature Type",
    labels = c("Tmax" = "Maximum", "Tavg" = "Average", "Tmin" = "Minimum")
  ) +
  labs(
    title = "Decadal Temperature Trends (1936-2024)",
    subtitle = "Hamburg Fuhlsbüttel",
    x = "",
    y = "Temperature (°C)"
  ) +
  theme_minimal(base_size = 16) +
  theme(
    plot.title = element_text(face = "bold", size = 24),
    plot.subtitle = element_text(size = 20),
    axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", color = "black", size = 20),
    axis.text.y = element_text(face = "bold"),
    legend.title = element_text(face = "bold"),
    legend.text = element_text(face = "bold"),
    legend.position = "none"

  ) +

# Add custom legend inside the plot
annotate("point", x = 2.5, y = 13.5, size = 5, shape = 21, fill = "#ff1744") +
annotate("text", x = 2.6, y = 13.5, label = "Maximum Temperature", hjust = 0, size = 6, fontface = "bol

annotate("point", x = 2.5, y = 9.5, size = 5, shape = 21, fill = "#fc8d59") +
annotate("text", x = 2.6, y = 9.5, label = "Average Temperature", hjust = 0, size = 6, fontface = "bold

annotate("point", x = 2.5, y = 6, size = 5, shape = 21, fill = "#4575b4") +
annotate("text", x = 2.6, y = 6, label = "Minimum Temperature", hjust = 0, size = 6, fontface = "bold")
```



Incremental changes in Temperature across Decades

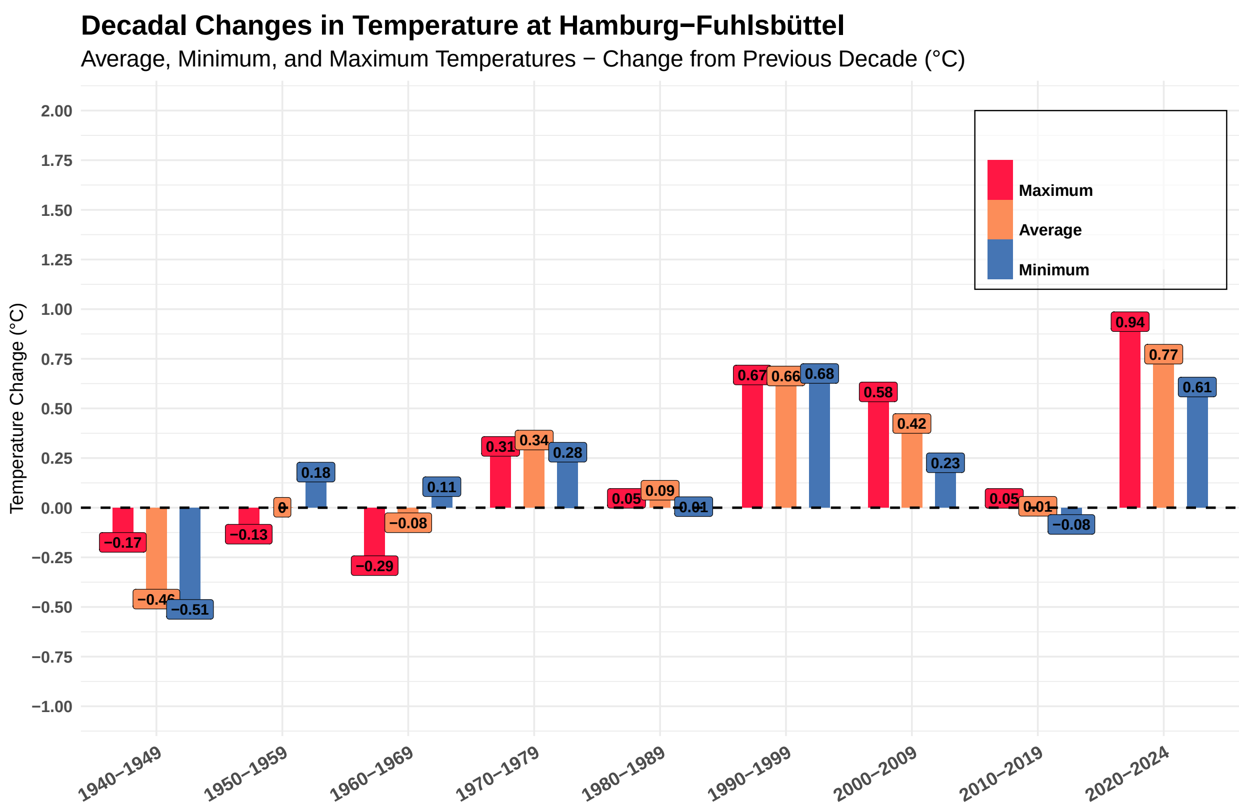
```
decadal_diff_tbl <- decadal_summary_tbl %>%
  arrange(Temperature_Type, decade_group) %>%
  group_by(Temperature_Type) %>%
  mutate(decadal_diff = decadal_mean - lag(decadal_mean)) %>%
  filter(!is.na(decadal_diff)) %>%
  ungroup()

ggplot(decadal_diff_tbl, aes(x = decade_group, y = decadal_diff, fill = Temperature_Type)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_label(aes(label = round(decadal_diff, 2)),
             position = position_dodge(width = 0.7),
             size = 4.5, fontface = "bold",
             show.legend = FALSE) +
  # geom_text(aes(label = round(decadal_diff, 2)),
  #           position = position_dodge(width = 0.7),
  #           vjust = ifelse(decadal_diff_tbl$decadal_diff >= 0, -0.3, 1.3),
  #           size = 5, fontface = "bold") +
  scale_fill_manual(
    values = c("Tmax" = "#ff1744", "Tavg" = "#fc8d59", "Tmin" = "#4575b4"),
    name = "Temperature Type",
    labels = c("Tmax" = "Maximum", "Tavg" = "Average", "Tmin" = "Minimum")
  ) +
  labs(
```

```r
  title = "Decadal Changes in Temperature at Hamburg-Fuhlsbüttel",
  subtitle = "Average, Minimum, and Maximum Temperatures - Change from Previous Decade (°C)"
) +

  theme_minimal(base_size = 16) +
  theme(
    plot.title = element_text(face = "bold", size = 24),
    plot.subtitle = element_text(size = 20),
    axis.text.x = element_text(angle = 30, hjust = 1),
    axis.text.y = element_text(face = "bold"),
    legend.title = element_text(face = "bold"),
    legend.text = element_text(face = "bold")
  )
```




```r
ggplot(decadal_diff_tbl, aes(x = decade_group, y = decadal_diff, fill = Temperature_Type)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.5) +
  geom_label(aes(label = round(decadal_diff, 2)),
             position = position_dodge(width = 0.8),
             size = 4.5, fontface = "bold",
             show.legend = FALSE) +
  geom_hline(yintercept = 0, color = "black", linewidth = 1, linetype = "dashed") +

  # Manual fill colors
  scale_fill_manual(
    values = c("Tmax" = "#ff1744", "Tavg" = "#fc8d59", "Tmin" = "#4575b4")
  ) +
```

```r
 scale_y_continuous(
  limits = c(-1, 2),
  breaks = seq(-2, 2, by = 0.25),
  labels = function(x) sprintf("%.2f", x)
) +
  # Titles
  labs(
    title = "Decadal Changes in Temperature at Hamburg-Fuhlsbüttel",
    subtitle = "Average, Minimum, and Maximum Temperatures - Change from Previous Decade (°C)",
    x = "", y = "Temperature Change (°C)"
  ) +

  # Theme & custom key
  theme_minimal(base_size = 16) +
  theme(
    plot.title = element_text(face = "bold", size = 24),
    plot.subtitle = element_text(size = 20),
    axis.text.x = element_text(angle = 30, hjust = 1, size = 16, face = "bold"),
    axis.text.y = element_text(face = "bold", size = 14),
    legend.position = "none"
  ) +

  # Custom legend inside plot

  # Custom legend inside plot for y-axis from -1 to 2
annotate("rect", xmin = 7.5, xmax = 9.5, ymin = 1.2, ymax = 2, fill = "white", alpha = 0.7) +  # backgr
annotate("rect", xmin = 7.5, xmax = 9.5, ymin = 1.1, ymax = 2, colour = "black", fill = NA) +  # border

annotate("rect", xmin = 7.6, xmax = 7.8, ymin = 1.55, ymax = 1.75, fill = "#ff1744") +
annotate("text", x = 7.85, y = 1.6, label = "Maximum", hjust = 0, size = 5, fontface = "bold") +

annotate("rect", xmin = 7.6, xmax = 7.8, ymin = 1.35, ymax = 1.55, fill = "#fc8d59") +
annotate("text", x = 7.85, y = 1.4, label = "Average", hjust = 0, size = 5, fontface = "bold") +

annotate("rect", xmin = 7.6, xmax = 7.8, ymin = 1.15, ymax = 1.35, fill = "#4575b4") +
annotate("text", x = 7.85, y = 1.2, label = "Minimum", hjust = 0, size = 5, fontface = "bold")
```




```r
library(ggplot2)
library(dplyr)
library(scales)
library(ggrepel)  # For repelling labels
```

```r
identify_heatwaves <- function(date, tmax, threshold, min_duration = 3) {
  # Ensure inputs are valid
  stopifnot(length(date) == length(tmax))
  stopifnot(is.numeric(tmax))
  stopifnot(inherits(date, "Date"))

  # Sort and prepare
  data <- data.frame(date = date, tmax = tmax)
  data <- data[order(data$date), ]
  data$above_threshold <- data$tmax > threshold

  # Initialize output
  heatwaves <- data.frame(
    start_date = as.Date(character()),
```

```r
  end_date = as.Date(character()),
  duration = integer(),
  avg_intensity = numeric(),
  total_intensity = numeric(),
  max_intensity = numeric(),
  mean_tmax = numeric(),
  peak_tmax = numeric(),
  stringsAsFactors = FALSE
)

# Tracking flags
in_heatwave <- FALSE
heatwave_start <- NULL
heatwave_length <- 0

# Loop through
for (i in seq_len(nrow(data))) {
  if (data$above_threshold[i]) {
    if (!in_heatwave) {
      in_heatwave <- TRUE
      heatwave_start <- data$date[i]
      heatwave_length <- 1
    } else {
      heatwave_length <- heatwave_length + 1
    }
  } else {
    if (in_heatwave && heatwave_length >= min_duration) {
      end_date <- data$date[i - 1]
      heatwave_days <- which(data$date >= heatwave_start & data$date <= end_date)
      exceedance <- data$tmax[heatwave_days] - threshold

      heatwaves <- rbind(heatwaves, data.frame(
        start_date = heatwave_start,
        end_date = end_date,
        duration = heatwave_length,
        avg_intensity = mean(exceedance),
        total_intensity = sum(exceedance),
        max_intensity = max(exceedance),
        mean_tmax = mean(data$tmax[heatwave_days]),
        peak_tmax = max(data$tmax[heatwave_days])
      ))
    }
    in_heatwave <- FALSE
    heatwave_length <- 0
  }
}

# Final check at end
if (in_heatwave && heatwave_length >= min_duration) {
  end_date <- data$date[nrow(data)]
  heatwave_days <- which(data$date >= heatwave_start & data$date <= end_date)
  exceedance <- data$tmax[heatwave_days] - threshold
```

```
    heatwaves <- rbind(heatwaves, data.frame(
      start_date = heatwave_start,
      end_date = end_date,
      duration = heatwave_length,
      avg_intensity = mean(exceedance),
      total_intensity = sum(exceedance),
      max_intensity = max(exceedance),
      mean_tmax = mean(data$tmax[heatwave_days]),
      peak_tmax = max(data$tmax[heatwave_days])
    ))
  }

  # Add year column
  if (nrow(heatwaves) > 0) {
    heatwaves$year <- as.numeric(format(heatwaves$start_date, "%Y"))
  }

  return(heatwaves)
}

heatwave_events_tbl <- identify_heatwaves(
  date = data$date,
  tmax = data$Tmax,
  threshold = 28,
  min_duration = 3
)

heatwave_count_per_year <- heatwave_events_tbl %>%
  count(year, name = "num_events")

heatwave_summary_tbl <- heatwave_events_tbl %>%
  group_by(year) %>%
  summarise(
    num_events = n(),
    total_duration = sum(duration, na.rm = TRUE),
    avg_duration = mean(duration, na.rm = TRUE),
    mean_avg_intensity = mean(avg_intensity, na.rm = TRUE),
    mean_max_intensity = mean(max_intensity, na.rm = TRUE),
    total_intensity = sum(total_intensity, na.rm = TRUE),
    max_peak_tmax = max(peak_tmax, na.rm = TRUE),
    .groups = "drop"
  )

# Convert date columns
 heatwave_events_tbl$start_date <- as.Date(heatwave_events_tbl$start_date)
heatwave_events_tbl$end_date <- as.Date(heatwave_events_tbl$end_date)

heatwave_events_tbl$year <- as.numeric(heatwave_events_tbl$year)
#
```

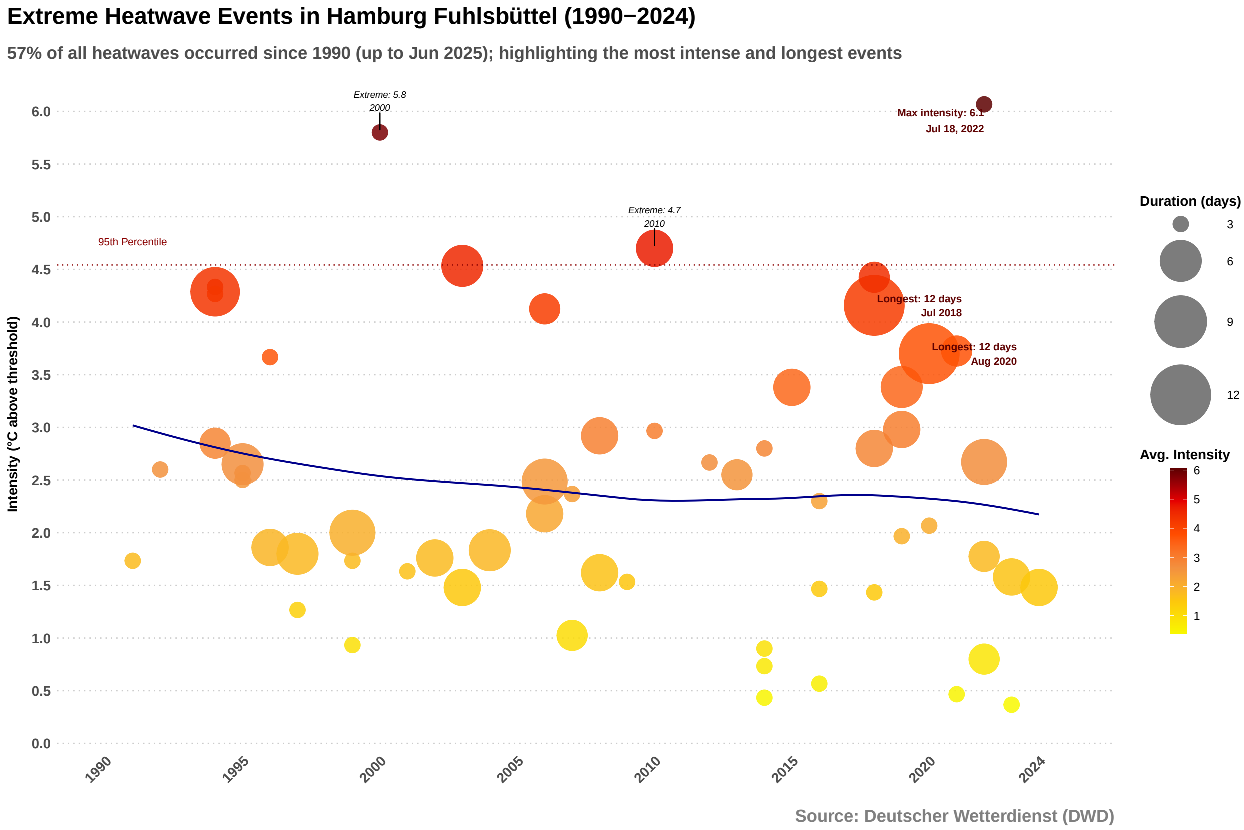
```
# Custom color palette
custom_colors <- c("#f8f800", "#fdc70c", "#f3903f", "#ff4d00", "#e60000", "#5e0000")
```

```r
# Filter events after 1990
heatwave_recent_tbl <- heatwave_events_tbl %>% filter(year >= 1990)

# Count and percentage
recent_events <- nrow(heatwave_recent_tbl)
total_events <- nrow(heatwave_events_tbl)
percent_recent <- round((recent_events / total_events) * 100)

# 95th percentile of avg_intensity in recent events
threshold_95 <- quantile(heatwave_recent_tbl$avg_intensity, 0.95, na.rm = TRUE)

# Max intensity among recent events
max_intensity_value <- max(heatwave_recent_tbl$avg_intensity, na.rm = TRUE)
longest_duration_value <- max(heatwave_recent_tbl$duration, na.rm = TRUE)

# Label most intense event(s)
max_intensity_event <- heatwave_recent_tbl %>%
  filter(avg_intensity == max_intensity_value) %>%
  mutate(label = paste0("Max intensity: ", round(avg_intensity, 1), "\n", format(start_date, "%b %d, %Y

# Label longest duration (among recent)
longest_duration_event <- heatwave_recent_tbl %>%
  filter(duration == longest_duration_value) %>%
  mutate(label = paste0("Longest: ", duration, " days\n", format(start_date, "%b %Y")))

# Extreme events = above 95th percentile (excluding the max to avoid duplicate label)
extreme_events <- heatwave_recent_tbl %>%
  filter(avg_intensity > threshold_95, avg_intensity < max_intensity_value) %>%
  mutate(label = paste0("Extreme: ", round(avg_intensity, 1), "\n", format(start_date, "%Y")))

# Plot
ggplot(heatwave_recent_tbl, aes(x = as.numeric(year), y = avg_intensity)) +
  geom_point(aes(size = duration, color = avg_intensity), alpha = 0.85) +

  geom_smooth(method = "loess", formula = y ~ x, se = FALSE,
              color = "darkblue", linetype = "solid", linewidth = 0.8, alpha = 0.6) +

  geom_hline(yintercept = threshold_95, linetype = "dotted", color = "darkred") +
  annotate("text", x = 1991, y = threshold_95 + 0.15,
         label = "95th Percentile",
         vjust = -0.5, color = "darkred", size = 3) +

  geom_text(data = max_intensity_event, aes(label = label),
            hjust = 1, vjust = 1.2, size = 3.3, fontface = "bold", color = "#5e0000") +

  ggrepel::geom_text_repel(data = longest_duration_event, aes(label = label),
                           size = 3, fontface = "bold", color = "#5e0000",
                           max.overlaps = 5, box.padding = 0.4, point.padding = 0.4,
                           hjust = 1) +

  ggrepel::geom_text_repel(data = extreme_events, aes(label = label),
    size = 2.8, fontface = "italic", color = "black",
    max.overlaps = 10, box.padding = 0.4, point.padding = 0.3,
```

```r
    nudge_y = 0.3) +

  scale_y_continuous(limits = c(0, 6.25), breaks = seq(0, 6.25, 0.5), expand = c(0, 0.05)) +
  scale_x_continuous(limits = c(1990, 2025), breaks = c(seq(1990, 2024, 5),2024) ) +

  scale_color_gradientn(colors = custom_colors,
                        name = "Avg. Intensity",
                        guide = guide_colorbar(barwidth = 1, barheight = 10,
                                               title.position = "top", title.hjust = 0.5)) +

  scale_size_continuous(
  range = c(6, 24),  # or any range that suits your plot size
  limits = c(3, 12), # ensures 3-12 maps to this range exactly
  breaks = c(3, 6, 9, 12),
  name = "Duration (days)"
) +

guides(
  size = guide_legend(
    title.position = "top",
    title.hjust = 0.5,
    override.aes = list(color = "gray40", fill = "gray40")  # Makes legend dots visible
  )
) +

  labs(
    title = "Extreme Heatwave Events in Hamburg Fuhlsbüttel (1990-2024)",
   subtitle = paste0(
  percent_recent, "% of all heatwaves occurred since 1990 (up to ",
  format(Sys.Date(), "%b %Y"), "); highlighting the most intense and longest events"
),

    x = "",
    y = "Intensity (°C above threshold)",
    caption = "Source: Deutscher Wetterdienst (DWD)"
  ) +

  theme_minimal(base_size = 12) +
  theme(
    plot.title.position = "plot",
    plot.title = element_text(face = "bold", size = 20, margin = margin(b = 15)),
    plot.subtitle = element_text(size = 15, color = "gray30", margin = margin(b = 15), face = "bold"),
    plot.caption = element_text(size = 15, color = "gray50",  face = "bold",hjust = 1),
    axis.title.y = element_text(margin = margin(r = 10), face = "bold", size = 12),
    axis.text.x  = element_text(angle = 45, hjust = 1, face = "bold", color = "gray30", size = 12),
    axis.text.y  = element_text(face = "bold", color = "gray30", size = 12),
    legend.position = "right",
    legend.box = "vertical",
    legend.margin = margin(l = 10),
    legend.title = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    panel.grid.major.y = element_line(color = "gray80", linetype = "dotted")
```

```
)
```

Heatwave Events :- We visualize the intensity and duration of heat events from 1936-2024.



Extreme Heatwave using Maximum Intensity

```r
# Filter post-1990 events
heatwave_recent_tbl <- heatwave_events_tbl %>%
  filter(year >= 1990)

# Total events
recent_events_max <- heatwave_recent_tbl %>%
  filter(max_intensity > 5) %>%
  summarize(count = n()) %>%
  pull(count)

percent_recent_max <- round((recent_events_max / nrow(heatwave_events_tbl)) * 100)

# 95th percentile of max_intensity among recent events
threshold_95_max <- quantile(heatwave_recent_tbl$max_intensity, 0.95, na.rm = TRUE)

# Maximum of max_intensity
max_max_intensity <- max(heatwave_recent_tbl$max_intensity, na.rm = TRUE)

# Max exceedance event
```

```r
max_max_event <- heatwave_recent_tbl %>%
  filter(max_intensity == max_max_intensity) %>%
  mutate(label = paste0("Max exceedance: ", round(max_intensity, 1), "\n", format(start_date, "%b %d, %Y

# Extreme events (excluding absolute max)
extreme_events_max <- heatwave_recent_tbl %>%
  filter(max_intensity > threshold_95_max, max_intensity < max_max_intensity) %>%
  mutate(label = paste0("Extreme: ", round(max_intensity, 1), "\n", format(start_date, "%Y")))


# Plot for Max Intensity (Post-1990)
ggplot(heatwave_recent_tbl, aes(x = as.numeric(year), y = max_intensity)) +

  geom_point(aes(size = duration, color = max_intensity), alpha = 0.85) +

  geom_smooth(method = "loess", formula = y ~ x, se = FALSE,
              color = "darkblue", linetype = "solid", linewidth = 0.8, alpha = 0.6) +

  geom_hline(yintercept = threshold_95_max, linetype = "dotted", color = "darkred") +
  annotate("text", x = 1990.5, y = threshold_95_max + 0.4, label = "95th Percentile",
           color = "darkred", size = 3) +

  geom_text(data = max_max_event, aes(label = label),
            hjust = 1, vjust = 1.2, size = 3.3, fontface = "bold", color = "#5e0000") +

  ggrepel::geom_text_repel(data = extreme_events_max, aes(label = label),
                           size = 2.8, fontface = "italic", color = "black",
                           max.overlaps = 10, box.padding = 0.4, point.padding = 0.3) +

  scale_y_continuous(limits = c(0, 12), breaks = seq(0, 12, 1), expand = c(0, 0.05)) +
  scale_x_continuous(limits = c(1990, 2024), breaks = c(seq(1990, 2024, 5),2024)) +

  scale_color_gradientn(colors = custom_colors,
                        name = "Max Intensity",
                        guide = guide_colorbar(barwidth = 1.5, barheight = 14,
                                               title.position = "top", title.hjust = 0.5)) +

   scale_size_continuous(
  range = c(6, 24),  # or any range that suits your plot size
  limits = c(3, 12), # ensures 3-12 maps to this range exactly
  breaks = c(3, 6, 9, 12),
  name = "Duration (days)"
) +

guides(
  size = guide_legend(
    title.position = "top",
    title.hjust = 0.5,
    override.aes = list(color = "gray40", fill = "gray40")  # Makes legend dots visible
  )
) +
  labs(
    title = "Max Daily Exceedance in Hamburg Fuhlsbüttel Heatwave Events (1990-2024)",
```

```r
  subtitle = paste0("Based on max daily exceedance above 28°C threshold\n",
                    percent_recent_max, "% of all events occurred since 1990"),
  x = "",
  y = "Max Intensity (°C above threshold)",
  caption = "Source: Deutscher Wetterdienst (DWD)"
) +

theme_minimal(base_size = 12) +
theme(
  plot.title.position = "plot",
  plot.title = element_text(face = "bold", size = 22, margin = margin(b = 15)),
  plot.subtitle = element_text(size = 15, color = "gray30", margin = margin(b = 15), face = "bold"),
  plot.caption = element_text(size = 12, color = "gray50",  face = "bold", hjust = 1),

  axis.title.y = element_text(margin = margin(r = 10), face = "bold", size = 16),
  axis.text.x  = element_text(angle = 45, hjust = 1, face = "bold", color = "gray30", size = 14),
  axis.text.y  = element_text(face = "bold", color = "gray30", size = 14),
  legend.position = "right",
  legend.box = "vertical",
  legend.margin = margin(l = 10),
  legend.title = element_text(face = "bold"),
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_blank(),
  panel.grid.major.y = element_line(color = "gray80", linetype = "dotted")
)
```